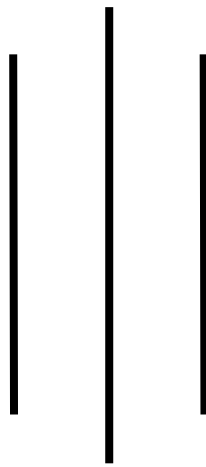


TRIBHUVAN UNIVERSITY

PATAN MULTIPLE CAMPUS

PATANDHOKA, LALITPUR



DATABASE ADMINISTRATION (BIT 352)

SUBMITTED BY:

NAME: SURESH DAHAL

ROLL: 23

CLASS: BIT-III/II

DATE: 2083/04/22

SUBMITTED TO:

BIPIN TIMALSINA

CHECKED BY

11. Working With RMAN Backup and Recovery

OUTPUT

a) Preparation and Configuration for Backup

Step 1: Connect as SYSDBA

```
C:\Users\VICTUS>sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sat Aug 1 07:54:38 2026
Version 19.3.0.0.0

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```

Step 2: Check whether the database is in ARCHIVELOG mode

```
Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

SQL> ARCHIVE LOG LIST;
Database log mode                No Archive Mode
Automatic archival                Disabled
Archive destination              C:\Oracle_19C\RDBMS
Oldest online log sequence       32
Current log sequence             34
```

Step 3: Enable ARCHIVELOG mode

```
SQL> SHUTDOWN IMMEDIATE;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL>
SQL> STARTUP MOUNT;
ORACLE instance started.

Total System Global Area 4915720712 bytes
Fixed Size                  9277960 bytes
Variable Size               989855744 bytes
Database Buffers           3909091328 bytes
Redo Buffers                 7495680 bytes
Database mounted.
SQL>
SQL> ALTER DATABASE ARCHIVELOG;

Database altered.

SQL>
SQL> ALTER DATABASE OPEN;

Database altered.
```

Step 4: Check Fast Recovery Area (FRA)

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SHOW PARAMETER db_recovery_file_dest;

NAME                                TYPE                                VALUE
-----                                -                                -
db_recovery_file_dest                string                              C:\Oracle_19C\FRA
db_recovery_file_dest_size            big integer                          0
```

Step 5: Connect to RMAN

```
C:\Users\VICTUS>rman target /

Recovery Manager: Release 19.0.0.0.0 - Production on Sat Aug 1 08:04:16 2026
Version 19.3.0.0.0

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connected to target database: ORCL (DBID=1764205357)
```

Step 6: Display current RMAN configuration

```
RMAN> show all;

using target database control file instead of recovery catalog
RMAN configuration parameters for database with db_unique_name ORCL are:
CONFIGURE RETENTION POLICY TO REDUNDANCY 1; # default
CONFIGURE BACKUP OPTIMIZATION OFF; # default
CONFIGURE DEFAULT DEVICE TYPE TO DISK; # default
CONFIGURE CONTROLFILE AUTOBACKUP ON; # default
CONFIGURE CONTROLFILE AUTOBACKUP FORMAT FOR DEVICE TYPE DISK TO '%F'; # default
CONFIGURE DEVICE TYPE DISK PARALLELISM 1 BACKUP TYPE TO BACKUPSET; # default
CONFIGURE DATAFILE BACKUP COPIES FOR DEVICE TYPE DISK TO 1; # default
CONFIGURE ARCHIVELOG BACKUP COPIES FOR DEVICE TYPE DISK TO 1; # default
CONFIGURE MAXSETSIZE TO UNLIMITED; # default
CONFIGURE ENCRYPTION FOR DATABASE OFF; # default
CONFIGURE ENCRYPTION ALGORITHM 'AES128'; # default
CONFIGURE COMPRESSION ALGORITHM 'BASIC' AS OF RELEASE 'DEFAULT' OPTIMIZE FOR LOAD TRUE ; # default
CONFIGURE RMAN OUTPUT TO KEEP FOR 7 DAYS; # default
CONFIGURE ARCHIVELOG DELETION POLICY TO NONE; # default
CONFIGURE SNAPSHOT CONTROLFILE NAME TO 'C:\ORACLE_19C\DATABASE\SNCFORCL.ORA'; # default
```

Step 7: Enable Control File Autobackup

```
RMAN> CONFIGURE CONTROLFILE AUTOBACKUP ON;

new RMAN configuration parameters:
CONFIGURE CONTROLFILE AUTOBACKUP ON;
new RMAN configuration parameters are successfully stored
```

```
RMAN> show all;

RMAN configuration parameters for database with db_unique_name ORCL are:
CONFIGURE RETENTION POLICY TO REDUNDANCY 1; # default
CONFIGURE BACKUP OPTIMIZATION OFF; # default
CONFIGURE DEFAULT DEVICE TYPE TO DISK; # default
CONFIGURE CONTROLFILE AUTOBACKUP ON;
CONFIGURE CONTROLFILE AUTOBACKUP FORMAT FOR DEVICE TYPE DISK TO '%F'; # default
CONFIGURE DEVICE TYPE DISK PARALLELISM 1 BACKUP TYPE TO BACKUPSET; # default
CONFIGURE DATAFILE BACKUP COPIES FOR DEVICE TYPE DISK TO 1; # default
CONFIGURE ARCHIVELOG BACKUP COPIES FOR DEVICE TYPE DISK TO 1; # default
CONFIGURE MAXSETSIZE TO UNLIMITED; # default
CONFIGURE ENCRYPTION FOR DATABASE OFF; # default
CONFIGURE ENCRYPTION ALGORITHM 'AES128'; # default
CONFIGURE COMPRESSION ALGORITHM 'BASIC' AS OF RELEASE 'DEFAULT' OPTIMIZE FOR LOAD TRUE ; # default
CONFIGURE RMAN OUTPUT TO KEEP FOR 7 DAYS; # default
CONFIGURE ARCHIVELOG DELETION POLICY TO NONE; # default
CONFIGURE SNAPSHOT CONTROLFILE NAME TO 'C:\ORACLE_19C\DATABASE\SNCFORCL.ORA'; # default
```

b) Perform Full Backup

Step 1: Backup the entire database including archived redo logs

```
RMAN> BACKUP DATABASE PLUS ARCHIVELOG;

Starting backup at 01-AUG-26
current log archived
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=1351 device type=DISK
channel ORA_DISK_1: starting archived log backup set
channel ORA_DISK_1: specifying archived log(s) in backup set
input archived log thread=1 sequence=34 RECID=1 STAMP=1240128452
channel ORA_DISK_1: starting piece 1 at 01-AUG-26
channel ORA_DISK_1: finished piece 1 at 01-AUG-26
piece handle=C:\ORACLE_19C\DATABASE\014ULMU5_1_1 tag=TAG20260801T080733 comment=NONE
channel ORA_DISK_1: backup set complete, elapsed time: 00:00:01
Finished backup at 01-AUG-26
```

Step 2: Verify backup

```
RMAN> list backup;

List of Backup Sets
=====

BS Key   Size                Device Type Elapsed Time Completion Time
-----
1        118.56M         DISK            00:00:00      01-AUG-26
BP Key: 1  Status: AVAILABLE Compressed: NO  Tag: TAG20260801T080733
Piece Name: C:\ORACLE_19C\DATABASE\014ULMU5_1_1

List of Archived Logs in backup set 1
Thrd Seq   Low SCN   Low Time   Next SCN   Next Time
----
1     34      4723890   31-JUL-26 4766775    01-AUG-26

BS Key   Type LV Size           Device Type Elapsed Time Completion Time
-----
2        Full  1.49G         DISK            00:00:02      01-AUG-26
BP Key: 2  Status: AVAILABLE Compressed: NO  Tag: TAG20260801T080734
Piece Name: C:\ORACLE_19C\DATABASE\024ULMU6_1_1

List of Datafiles in backup set 2
File LV Type Ckp SCN   Ckp Time   Abs Fuz SCN Sparse Name
----
1     Full 4766792   01-AUG-26          NO C:\ORACLE19C\ORADATA\ORCL\SYSTEM01.DBF
3     Full 4766792   01-AUG-26          NO C:\ORACLE19C\ORADATA\ORCL\SYS_AUX01.DBF
4     Full 4766792   01-AUG-26          NO C:\ORACLE19C\ORADATA\ORCL\UNDOTBS01.DBF
```

c) Simulate Database Failure

Step 1: Find the USERS datafile

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> select file#, name from v$datafile;

FILE# NAME
-----
1 C:\ORACLE19C\ORADATA\ORCL\SYSTEM01.DBF
3 C:\ORACLE19C\ORADATA\ORCL\SYS_AUX01.DBF
4 C:\ORACLE19C\ORADATA\ORCL\UNDOTBS01.DBF
5 C:\ORACLE19C\ORADATA\ORCL\PDBSEED\SYSTEM01.DBF
6 C:\ORACLE19C\ORADATA\ORCL\PDBSEED\SYS_AUX01.DBF
7 C:\ORACLE19C\ORADATA\ORCL\USERS01.DBF
8 C:\ORACLE19C\ORADATA\ORCL\PDBSEED\UNDOTBS01.DBF
9 C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SYSTEM01.DBF
10 C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SYS_AUX01.DBF
11 C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\UNDOTBS01.DBF
12 C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\USERS01.DBF
```

Step 2: Shutdown database

```
SQL> SHUTDOWN IMMEDIATE;  
Database closed.  
Database dismounted.  
ORACLE instance shut down.
```

Step 3: Delete the USERS datafile

```
C:\Oracle19C\oradata\ORCL\orclpdb>del C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SU  
RESH_DF2_23.DBF
```

Step 4: Switch to OCRLPDB and open database

```
SQL> alter database open  
2 ;  
alter database open  
*  
ERROR at line 1:  
ORA-01157: cannot identify/lock data file 14 - see DBWR trace file  
ORA-01110: data file 14: 'C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF2_23.  
DBF'
```

d) Restore and Recover the Datafile

Step 1: Connect to RMAN

rman target /

```
C:\Oracle19C\oradata\ORCL\orclpdb>rman target /  
  
Recovery Manager: Release 19.0.0.0.0 - Production on Sat Aug 1 10:30:28 20  
26  
Version 19.3.0.0.0  
  
Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights reserv  
ed.  
  
connected to target database: ORCL (DBID=1764205357)  
  
RMAN> |
```

Step 2: Restore the datafile

```
RMAN> restore datafile 14;  
  
Starting restore at 01-AUG-26  
allocated channel: ORA_DISK_1  
channel ORA_DISK_1: SID=983 device type=DISK  
  
channel ORA_DISK_1: starting datafile backup set restore  
channel ORA_DISK_1: specifying datafile(s) to restore from backup set  
channel ORA_DISK_1: restoring datafile 00014 to C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF2_23.DBF  
channel ORA_DISK_1: reading from backup piece C:\ORACLE_19C\DATABASE\034ULMU9_1_1  
channel ORA_DISK_1: piece handle=C:\ORACLE_19C\DATABASE\034ULMU9_1_1 tag=TAG20260801T080734  
channel ORA_DISK_1: restored backup piece 1  
channel ORA_DISK_1: restore complete, elapsed time: 00:00:01  
Finished restore at 01-AUG-26
```

Step 3: Recover the datafile

```
RMAN> recover datafile 14;

Starting recover at 01-AUG-26
using channel ORA_DISK_1

starting media recovery
media recovery complete, elapsed time: 00:00:00

Finished recover at 01-AUG-26
```

Step 4: Verify

```
Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

SQL> select file#, status from v$datafile where file#=14;

  FILE# STATUS
  -----
     14 ONLINE
```

e) Full Database Recovery Simulation

Step 1: Shutdown database

```
SQL> exec my_info;
Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

SQL> shutdown immediate;
Pluggable Database closed.
SQL> |
```

Step 2: Delete the SPFILE, control files and datafiles

```
C:\>del "C:\Oracle_19C\database\SPFILEORCL.ORA"

C:\>del C:\ORACLE19C\ORADATA\ORCL\CONTROL01.CTL

C:\>del C:\ORACLE19C\ORADATA\ORCL\CONTROL02.CTL

C:\>del C:\ORACLE19C\ORADATA\ORCL\*.dbf

C:\>|
```

Step 3: Connect RMAN

```
C:\Users\VICTUS>rman target /

Recovery Manager: Release 19.0.0.0.0 - Production on Sat Aug 1 12:11:10 2026
Version 19.3.0.0.0

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connected to target database: ORCL (not mounted)
```

Step 4: Restore SPFILE

```
RMAN> RESTORE SPFILE FROM AUTOBACKUP;

Starting restore at 01-AUG-26
using target database control file instead of recovery catalog
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=539 device type=DISK

channel ORA_DISK_1: looking for AUTOBACKUP on day: 20260801
channel ORA_DISK_1: AUTOBACKUP found: c-1764205357-20260801-00
channel ORA_DISK_1: restoring spfile from AUTOBACKUP c-1764205357-20260801-00
channel ORA_DISK_1: SPFILE restore from AUTOBACKUP complete
Finished restore at 01-AUG-26
```

Step 5: Restore control file

```
RMAN> RESTORE CONTROLFILE FROM AUTOBACKUP;

Starting restore at 01-AUG-26
using channel ORA_DISK_1

channel ORA_DISK_1: looking for AUTOBACKUP on day: 20260801
channel ORA_DISK_1: AUTOBACKUP found: c-1764205357-20260801-00
channel ORA_DISK_1: restoring control file from AUTOBACKUP c-1764205357-20260801-00
channel ORA_DISK_1: control file restore from AUTOBACKUP complete
output file name=C:\ORACLE19C\ORADATA\ORCL\CONTROL01.CTL
output file name=C:\ORACLE19C\ORADATA\ORCL\CONTROL02.CTL
Finished restore at 01-AUG-26
```

Step 6: Restore database

```
RMAN> RESTORE DATABASE;

Starting restore at 01-AUG-26
using channel ORA_DISK_1

skipping datafile 5; already restored to file C:\ORACLE19C\ORADATA\ORCL\PDBSEED\SYSTEM01.DBF
skipping datafile 6; already restored to file C:\ORACLE19C\ORADATA\ORCL\PDBSEED\SYS_AUX01.DBF
skipping datafile 8; already restored to file C:\ORACLE19C\ORADATA\ORCL\PDBSEED\UNDOTBS01.DBF
channel ORA_DISK_1: starting datafile backup set restore
channel ORA_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_DISK_1: restoring datafile 00001 to C:\ORACLE19C\ORADATA\ORCL\SYSTEM01.DBF
channel ORA_DISK_1: restoring datafile 00003 to C:\ORACLE19C\ORADATA\ORCL\SYS_AUX01.DBF
channel ORA_DISK_1: restoring datafile 00004 to C:\ORACLE19C\ORADATA\ORCL\UNDOTBS01.DBF
channel ORA_DISK_1: restoring datafile 00007 to C:\ORACLE19C\ORADATA\ORCL\USERS01.DBF
channel ORA_DISK_1: reading from backup piece C:\ORACLE_19C\DATABASE\024ULMU6_1_1
channel ORA_DISK_1: piece handle=C:\ORACLE_19C\DATABASE\024ULMU6_1_1 tag=TAG20260801T080734
channel ORA_DISK_1: restored backup piece 1
channel ORA_DISK_1: restore complete, elapsed time: 00:00:03
channel ORA_DISK_1: starting datafile backup set restore
channel ORA_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_DISK_1: restoring datafile 00009 to C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SYSTEM01.DBF
channel ORA_DISK_1: restoring datafile 00010 to C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SYS_AUX01.DBF
channel ORA_DISK_1: restoring datafile 00011 to C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\UNDOTBS01.DBF
channel ORA_DISK_1: restoring datafile 00012 to C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\USERS01.DBF
channel ORA_DISK_1: restoring datafile 00013 to C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF1_23.DBF
channel ORA_DISK_1: restoring datafile 00014 to C:\ORACLE19C\ORADATA\ORCL\ORCLPDB\SURESH_DF2_23.DBF
channel ORA_DISK_1: reading from backup piece C:\ORACLE_19C\DATABASE\034ULMU9_1_1
channel ORA_DISK_1: piece handle=C:\ORACLE_19C\DATABASE\034ULMU9_1_1 tag=TAG20260801T080734
channel ORA_DISK_1: restored backup piece 1
channel ORA_DISK_1: restore complete, elapsed time: 00:00:04
Finished restore at 01-AUG-26
```

Step 7: Recover database

```
RMAN> recover database;

Starting recover at 01-AUG-26
using channel ORA_DISK_1

starting media recovery

archived log for thread 1 with sequence 35 is already on disk as file C:\ORACLE19C\ORADATA\ORCL\REDO02.LOG
archived log for thread 1 with sequence 36 is already on disk as file C:\ORACLE19C\ORADATA\ORCL\REDO03.LOG
archived log file name=C:\ORACLE19C\ORADATA\ORCL\REDO02.LOG thread=1 sequence=35
archived log file name=C:\ORACLE19C\ORADATA\ORCL\REDO03.LOG thread=1 sequence=36
media recovery complete, elapsed time: 00:00:01
Finished recover at 01-AUG-26
```

Step 8: Open database

```
RMAN> ALTER DATABASE OPEN RESETLOGS;  
  
Statement processed
```

f) Validation and Cleanup

Step 1: Validate database

```
RMAN> validate database;  
  
Starting validate at 01-AUG-26  
using channel ORA_DISK_1  
channel ORA_DISK_1: starting validation of datafile  
channel ORA_DISK_1: specifying datafile(s) for validation  
input datafile file number=00001 name=C:\ORACLE19C\ORADATA\ORCL\SYSTEM01.DBF  
input datafile file number=00003 name=C:\ORACLE19C\ORADATA\ORCL\SYS_AUX01.DBF  
input datafile file number=00004 name=C:\ORACLE19C\ORADATA\ORCL\UNDOTBS01.DBF  
input datafile file number=00007 name=C:\ORACLE19C\ORADATA\ORCL\USERS01.DBF  
channel ORA_DISK_1: validation complete, elapsed time: 00:00:01  
List of Datafiles  
=====
```

File	Status	Marked Corrupt	Empty Blocks	Blocks Examined	High SCN
1	OK	0	16718	116481	4788659
File Name: C:\ORACLE19C\ORADATA\ORCL\SYSTEM01.DBF					
Block Type Blocks Failing Blocks Processed					

Data	0		81347		
Index	0		13330		
Other	0		5085		

Step 2: Check corruption

```
RMAN> SELECT * FROM V$DATABASE_BLOCK_CORRUPTION;  
  
no rows selected
```

Step 3: Verify data

```
PL/SQL procedure successfully completed.  
  
SQL> SELECT username FROM dba_users;  
  
USERNAME  
-----  
SYS  
SYSTEM  
XS$NULL  
OJVM SYS  
LBAC SYS  
OUTLN  
SYS$UMF  
DBSNMP  
APPQOSSYS  
DBSFUSER  
GGSYS
```

12. LEARNING ABOUT STANDARD AUDITING IN ORACLE

OUTPUT

(a) Preparation and Configuration for Standard Auditing

```
Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

SQL> CONNECT / AS SYSDBA;
Connected.
SQL>
SQL> ALTER SESSION SET CONTAINER = ORCLPDB;

Session altered.

SQL>
SQL> SHOW PARAMETER audit_trail;

NAME                                TYPE        VALUE
-----
audit_trail                          string      DB
```

(b) Enable Auditing

```
Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

SQL> AUDIT SELECT, INSERT, UPDATE, DELETE
2  ON HR.EMPLOYEES
3  BY ACCESS;

Audit succeeded.
```

(c) Generate Audit Records

```
SQL> connect hr/thaxaena@orclpdb
Connected.
```

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> SELECT first_name, email, salary
2  FROM EMPLOYEES
3  WHERE EMPLOYEE_ID = 100;

FIRST_NAME      EMAIL              SALARY
-----
Steven          SKING              24000
```

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> UPDATE EMPLOYEES
2  SET SALARY = SALARY + 100
3  WHERE EMPLOYEE_ID = 100;

1 row updated.

SQL> commit;

Commit complete.
```

(d) Query and Analyze Audit Trail

```
SQL> connect / as sysdba;
Connected.
SQL> ALTER SESSION SET CONTAINER = ORCLPDB;

Session altered.

SQL> set serveroutput on;
SQL> exec my_info;
Name: Suresh Dahal \ Roll: 23
```

Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

```
SQL> SELECT username,
2         action_name,
3         obj_name,
4         timestamp,
5         sql_text
6 FROM dba_audit_trail
7 WHERE username='HR'
8 AND obj_name='EMPLOYEES'
9 ORDER BY timestamp DESC;
```

USERNAME	ACTION_NAME	OBJ_NAME	TIMESTAMP	SQL_TEXT
HR	UPDATE	EMPLOYEES	01-AUG-26	
HR	UPDATE	EMPLOYEES	01-AUG-26	
HR	SELECT	EMPLOYEES	01-AUG-26	
HR	SELECT	EMPLOYEES	01-AUG-26	
HR	SELECT	EMPLOYEES	01-AUG-26	
HR	SELECT	EMPLOYEES	01-AUG-26	
HR	SELECT	EMPLOYEES	01-AUG-26	

(e) Disable Auditing and Purge Audit Trail

Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

```
SQL> NOAUDIT SELECT, INSERT, UPDATE, DELETE
2 ON HR.EMPLOYEES;
```

Noaudit succeeded.

Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

```
SQL> BEGIN
2   DBMS_AUDIT_MGMT.CLEAN_AUDIT_TRAIL(
3     AUDIT_TRAIL_TYPE => DBMS_AUDIT_MGMT.AUDIT_TRAIL_AUD_STD,
4     USE_LAST_ARCH_TIMESTAMP => FALSE);
5 END;
6 /
```

PL/SQL procedure successfully completed.

Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

```
SQL> BEGIN
2   DBMS_AUDIT_MGMT.INIT_CLEANUP(
3     AUDIT_TRAIL_TYPE => DBMS_AUDIT_MGMT.AUDIT_TRAIL_AUD_STD,
4     DEFAULT_CLEANUP_INTERVAL => 24);
5 END;
6 /
```

(f) Validation

```
Name: Suresh Dahal \ Roll: 23
```

```
PL/SQL procedure successfully completed.
```

```
SQL> connect hr/thaxaena@orclpdb;  
Connected.
```

```
Name: Suresh Dahal \ Roll: 23
```

```
PL/SQL procedure successfully completed.
```

```
SQL>
```

```
SQL> SELECT username,  
2         action_name,  
3         obj_name,  
4         timestamp  
5 FROM dba_audit_trail  
6 WHERE username='HR'  
7 AND obj_name='EMPLOYEES'  
8 ORDER BY timestamp DESC;
```

```
no rows selected
```

13. WORKING WITH UNIFIED AUDITING IN ORACLE

OUTPUT

(a) Preparation and Verification

```
SQL> show con_name;

CON_NAME
-----
ORCLPDB

Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

SQL> SELECT parameter, value
2 FROM v$option
3 WHERE parameter = 'Unified Auditing';

PARAMETER
-----
VALUE
-----
Unified Auditing
FALSE
```

(b) Create and Enable Audit Policy

```
SQL> exec my_info;
Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

SQL> CREATE AUDIT POLICY hr_emp_policy
2 ACTIONS
3 SELECT ON HR.EMPLOYEES,
4 INSERT ON HR.EMPLOYEES,
5 UPDATE ON HR.EMPLOYEES,
6 DELETE ON HR.EMPLOYEES;

Audit policy created.
```

Enable the policy:

```
Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

SQL> AUDIT POLICY hr_emp_policy;

Audit succeeded.
```

(c) Generate Audit Records

Connect as HR:

```
SQL> connect hr/thaxaena@orclpdb
Connected.
```

Insert a record:

```
SQL> INSERT INTO employees
2 (employee_id, first_name, last_name, email, hire_date, job_id, salary) VALUES (999, 'Test', 'User', 'test@ex
ample.com', SYSDATE, 'IT_PROG', 5000);

1 row created.

SQL> commit;

Commit complete.
```

Delete the record:

```
Name: Suresh Dahal / Roll: 23/ Section: A
PL/SQL procedure successfully completed.
SQL> DELETE FROM employees
  2  WHERE employee_id = 999;
1 row deleted.
SQL> commit;
Commit complete.
```

(d) Query and Analyze Unified Audit Trail

```
SQL> connect / as sysdba;
Connected.
SQL> alter session set container = orclpdb;
Session altered.
```

Query the audit trail:

```
Name: Suresh Dahal \ Roll: 23
PL/SQL procedure successfully completed.
SQL> SELECT
  2     dbusername,
  3     action_name,
  4     object_schema,
  5     object_name,
  6     event_timestamp,
  7     sql_text
  8 FROM unified_audit_trail
  9 WHERE dbusername='HR'
 10 AND object_name='EMPLOYEES'
 11 ORDER BY event_timestamp DESC;
```

USER	ACTION	SCHEMA	OBJECT	TIMESTAMP	SQL QUERY
HR	DELETE	HR	EMPLOYEES	01-AUG-26 01.44.31.591000 PM	DELETE FROM employees WHERE employee_id = 999
HR	INSERT	HR	EMPLOYEES	01-AUG-26 01.43.41.331000 PM	INSERT INTO employees (employee_id, first_name, last_

name, email, hire_date, job

(e) Conditional Auditing

Create a conditional audit policy:

```
Name: Suresh Dahal \ Roll: 23
PL/SQL procedure successfully completed.
SQL> CREATE AUDIT POLICY hr_hr_user_pol
  2 ACTIONS UPDATE ON HR.EMPLOYEES
  3 WHEN 'SYS_CONTEXT(''USERENV'', 'SESSION_USER') = 'HR''
  4 EVALUATE PER STATEMENT;
Audit policy created.
```

Enable it:

```
SQL> AUDIT POLICY hr_hr_user_pol;
Audit succeeded.
```

Connect as HR:

```
PS C:\Users\VICTUS> sqlplus hr/thaxaena@orclpdb;

SQL*Plus: Release 19.0.0.0.0 - Production on Sat Aug 1 13:54:08 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Last Successful login time: Sat Aug 01 2026 13:39:23 +05:45

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

Execute a qualifying update:

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> UPDATE employees
  2 SET salary = 12000
  3 WHERE employee_id = 100;

1 row updated.

SQL> commit
  2 ;

Commit complete.
```

Verify the audit record:

```
SQL> CONNECT / AS SYSDBA;
Connected.
SQL>
SQL> ALTER SESSION SET CONTAINER = ORCLPDB;

Session altered.
```

```
Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

SQL> SELECT
  2 dbusername,
  3 action_name,
  4 event_timestamp,
  5 sql_text
  6 FROM unified_audit_trail
  7 WHERE dbusername='HR'
  8 ORDER BY event_timestamp DESC;
```

USER	ACTION	TIMESTAMP	SQL TEXT
HR	UPDATE	01-AUG-26 01.54.39.217000 PM	UPDATE employees SET salary = 12000 WHERE employee_id = 100
HR	LOGON	01-AUG-26 01.53.13.093000 PM	
HR	LOGON	01-AUG-26 01.53.02.855000 PM	
HR	LOGON	01-AUG-26 01.52.10.614000 PM	
HR	DELETE	01-AUG-26 01.44.31.591000 PM	DELETE FROM employees WHERE employee_id = 999

(f) Disable, Drop, and Cleanup

Disable both policies:

```
Name: Suresh Dahal \ Roll: 23  
PL/SQL procedure successfully completed.  
SQL> NOAUDIT POLICY hr_emp_policy;  
Noaudit succeeded.  
SQL>  
SQL> NOAUDIT POLICY hr_hr_user_pol;  
Noaudit succeeded.
```

Drop them:

```
SQL> DROP AUDIT POLICY hr_emp_policy;  
Audit Policy dropped.  
SQL>  
SQL> DROP AUDIT POLICY hr_hr_user_pol;  
Audit Policy dropped.
```

Purge the Unified Audit Trail:

```
Name: Suresh Dahal \ Roll: 23  
PL/SQL procedure successfully completed.  
SQL> BEGIN  
2   DBMS_AUDIT_MGMT.CLEAN_AUDIT_TRAIL(  
3       AUDIT_TRAIL_TYPE => DBMS_AUDIT_MGMT.AUDIT_TRAIL_UNIFIED,  
4       USE_LAST_ARCH_TIMESTAMP => FALSE);  
5 END;  
6 /  
PL/SQL procedure successfully completed.
```

Validation:

Connect as HR:

```
PS C:\Users\VICTUS> sqlplus hr/thaxaena@orclpdb;  
  
SQL*Plus: Release 19.0.0.0.0 - Production on Sat Aug 1 14:00:17 2026  
Version 19.3.0.0.0  
  
Copyright (c) 1982, 2019, Oracle. All rights reserved.  
  
Last Successful login time: Sat Aug 01 2026 13:54:08 +05:45  
  
Connected to:  
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Version 19.3.0.0.0
```

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> SELECT *
      2 FROM employees
      3 WHERE employee_id = 100;
```

	ID	FIRST NAME	LAST NAME	EMAIL	PHONE	HIRE DATE	JOB ID
D							
-							
0	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES

Connect as SYS:

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> CONNECT / AS SYSDBA;
Connected.
SQL>
SQL> ALTER SESSION SET CONTAINER = ORCLPDB;

Session altered.
```

Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

```
SQL> SELECT
      2     dbusername,
      3     action_name,
      4     object_name,
      5     event_timestamp
      6 FROM unified_audit_trail
      7 WHERE dbusername='HR'
      8 ORDER BY event_timestamp DESC;
```

no rows selected

14. EXPLORING FLASHBACK TECHNOLOGY IN ORACLE.

OUTPUT

(a) Preparation and Configuration

Step 1: Connect as SYS

```
C:\Users\VICTUS>sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sat Aug 1 19:25:26 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

Step 2: Check UNDO_RETENTION

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SHOW PARAMETER undo_retention;

NAME                                TYPE          VALUE
-----
undo_retention                      integer       900
```

Step 3: Check Recycle Bin

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SHOW PARAMETER recyclebin;

NAME                                TYPE          VALUE
-----
recyclebin                          string        on
```

Step 4: Set UNDO_RETENTION (if less than 3600)

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> ALTER SYSTEM SET undo_retention=3600;

System altered.
```

Step 5: Verify the change

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SHOW PARAMETER undo_retention;

NAME                                TYPE          VALUE
-----
undo_retention                      integer       3600
```

Step 6: Verify ARCHIVELOG mode

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> ARCHIVE LOG LIST;
Database log mode          Archive Mode
Automatic archival         Enabled
Archive destination        C:\Oracle_19C\RDBMS
Oldest online log sequence 1
Next log sequence to archive 1
Current log sequence        1
```

(b) Flashback Query

Step 1: Connect as HR

```
SQL> connect hr/thaxaena@orclpdb;
Connected.
SQL> show user;
USER is "HR"
```

Step 2: Check current salary

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> SELECT employee_id,
2          first_name,
3          last_name,
4          salary
5 FROM employees
6 WHERE employee_id=103;

EMPLOYEE_ID FIRST_NAME      LAST_NAME      SALARY
-----
103 Alexander      Hunold          9000
```

Step 3: Record current timestamp

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> SELECT SYSTIMESTAMP FROM dual;

SYSTIMESTAMP
-----
01-AUG-26 07.31.44.542000 PM +05:45
```

Step 4: Increase salary

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> UPDATE employees
2 SET salary = salary + 1000
3 WHERE employee_id=103;

1 row updated.

SQL> commit;

Commit complete.
```

Step 5: Verify updated salary

```
Name: Suresh Dahal/ Roll: 23/ Section: A
PL/SQL procedure successfully completed.

SQL> SELECT employee_id,salary
  2   FROM employees
  3   WHERE employee_id=103;

EMPLOYEE_ID      SALARY
-----
          103      10000
```

Step 6: Flashback Query

```
Name: Suresh Dahal/ Roll: 23/ Section: A
PL/SQL procedure successfully completed.

SQL> SELECT employee_id,
  2          salary
  3   FROM employees
  4   AS OF TIMESTAMP
  5   TO_TIMESTAMP(
  6    '2026-08-01 07:31:44',
  7    'YYYY-MM-DD HH24:MI:SS')
  8   WHERE employee_id=103;

EMPLOYEE_ID      SALARY
-----
          103       9000
```

(c) Flashback Version Query

Step 1: Check salary.

```
Name: Suresh Dahal/ Roll: 23/ Section: A
PL/SQL procedure successfully completed.

SQL> SELECT employee_id,salary
  2   FROM employees
  3   WHERE employee_id=104;

EMPLOYEE_ID      SALARY
-----
          104       6000
```

Step 2: Increase salary by 500.

```
Name: Suresh Dahal/ Roll: 23/ Section: A
PL/SQL procedure successfully completed.

SQL> UPDATE employees
  2   SET salary=salary+500
  3   WHERE employee_id=104;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.
```

Step 3

```
Name: Suresh Dahal/ Roll: 23/ Section: A
PL/SQL procedure successfully completed.
SQL> SELECT SYSTIMESTAMP FROM dual;
SYSTIMESTAMP
-----
01-AUG-26 07.36.35.177000 PM +05:45
```

Step 4: Increase salary again.

```
Name: Suresh Dahal/ Roll: 23/ Section: A
PL/SQL procedure successfully completed.
SQL> UPDATE employees
  2 SET salary=salary+600
  3 WHERE employee_id=104;
1 row updated.
SQL>
SQL> COMMIT;
Commit complete.
```

Step 5: Display all versions.

```
Name: Suresh Dahal/ Roll: 23/ Section: A
PL/SQL procedure successfully completed.
SQL> SELECT versions_starttime,
  2     versions_endtime,
  3     versions_operation,
  4     salary
  5 FROM employees
  6 VERSIONS BETWEEN
  7 TIMESTAMP
  8 SYSTIMESTAMP-INTERVAL '10' MINUTE
  9 AND SYSTIMESTAMP
 10 WHERE employee_id=104;
START TIME          END TIME          OP      SALARY
-----
01-AUG-26 07.37.00 PM          U          7100
01-AUG-26 07.36.09 PM 01-AUG-26 07.37.00 PM U          6500
01-AUG-26 07.36.09 PM          6000
```

(d) Flashback Table

Step 1: Connect as sys

```
SQL> connect / as sysdba
Connected.
SQL> show user
USER is "SYS"
```

Step 2: Enable row movement.

```
SQL> ALTER TABLE hr.employees
  2 ENABLE ROW MOVEMENT;
Table altered.
```

Step 3: Connect back as HR.

```
SQL> connect hr/thaxaena@orclpdb;
Connected.
SQL> show user;
USER is "HR"
```

Step 4: Delete employee 105.

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> DELETE
  2 FROM employees
  3 WHERE employee_id=105;

1 row deleted.

SQL>
SQL> COMMIT;

Commit complete.
```

Step 5: Record timestamp.

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> SELECT SYSTIMESTAMP FROM dual;

SYSTIMESTAMP
-----
01-AUG-26 07.43.08.989000 PM +05:45
```

Step 6: Flashback table.

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> FLASHBACK TABLE employees
  2 TO TIMESTAMP
  3 TO_TIMESTAMP(
  4 '2026-08-01 07:43:08',
  5 'YYYY-MM-DD HH24:MI:SS');

Flashback complete.
```

Step 7: Verify.

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> select first_name from employees where employee_id = 105;

FIRST_NAME
-----
David
```

(e) Flashback Drop

Step 1: Drop table.

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> DROP TABLE locations CASCADE CONSTRAINTS;

Table dropped.
```

Step 2: Check the recycle bin.

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> SHOW RECYCLEBIN;
ORIGINAL NAME      RECYCLEBIN NAME      OBJECT TYPE  DROP TIME
-----
LOCATIONS          BIN$w2EcpuQpQ1mevr8pPMxaww==$0  TABLE      2026-08-01:19:49:48
```

Step 3: Restore table.

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> FLASHBACK TABLE locations
  2  TO BEFORE DROP;

Flashback complete.
```

Step 4: Verify.

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> SELECT *
  2  FROM locations
  3  WHERE location_id=1000;

LOCATION_ID STREET_ADDRESS      POSTAL_CODE  CITY      STATE_PROVINCE  CO
-----
      1000 1297 Via Cola di Rie 00989      Roma      -----
                                         IT
```

(f) Validation and Cleanup

Restore employee 103 salary

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> UPDATE employees
  2  SET salary=salary-1000
  3  WHERE employee_id=103;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.
```

Restore employee 104 salary

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> UPDATE employees
  2  SET salary=salary-1100
  3  WHERE employee_id=104;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.
```

Delete employee 105 again

```
Name: Suresh Dahal/ Roll: 23/ Section: A  
PL/SQL procedure successfully completed.  
  
SQL> DELETE  
2 FROM employees  
3 WHERE employee_id=105;  
  
1 row deleted.  
  
SQL>  
SQL> COMMIT;  
  
Commit complete.
```

Demonstrating ORA-08180

```
Name: Suresh Dahal/ Roll: 23/ Section: A  
PL/SQL procedure successfully completed.  
  
SQL> SELECT first_name, last_name  
2 FROM employees  
3 AS OF TIMESTAMP  
4 (SYSTIMESTAMP-INTERVAL '2' MONTH)  
5 WHERE employee_id=103;  
FROM employees  
*  
ERROR at line 2:  
ORA-08180: no snapshot found based on specified time
```

15. ORACLE FLASHBACK VERSION QUERY AND TRANSACTION QUERY

OUTPUT

(a) Preparation and Configuration

Step 1: Connect as SYSDBA

```
C:\Users\VICTUS>sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sat Aug 1 20:18:08 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> show user
USER is "SYS"
```

Step 2: Grant required privileges to HR

```
Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

SQL> GRANT SELECT ANY TRANSACTION TO hr;

Grant succeeded.

SQL>
SQL> GRANT FLASHBACK ANY TABLE TO hr;

Grant succeeded.

SQL>
SQL> GRANT SELECT ON v_$transaction TO hr;

Grant succeeded.
```

Step 3: Check UNDO retention

```
Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

SQL> SHOW PARAMETER undo_retention;

NAME                                TYPE          VALUE
-----
undo_retention                       integer       3600
```

Step 4: Increase UNDO retention (if less than 7200)

```
Name: Suresh Dahal \ Roll: 23

PL/SQL procedure successfully completed.

SQL> ALTER SYSTEM SET undo_retention=7200;

System altered.
```

Step 5: Verify the parameter

```
SQL> SHOW PARAMETER undo_retention;
```

NAME	TYPE	VALUE
undo_retention	integer	7200

(b) Flashback Version Query Setup and Execution

Step 1: Connect as HR

```
SQL> connect hr/thaxaena@orclpdb;  
Connected.
```

Step 2: View current salary

```
Name: Suresh Dahal/ Roll: 23/ Section: A  
PL/SQL procedure successfully completed.  
SQL> SELECT employee_id,  
2          salary  
3 FROM employees  
4 WHERE employee_id=106;  
  
EMPLOYEE_ID    SALARY  
-----  
106            4800
```

Step 3: First update

```
Name: Suresh Dahal/ Roll: 23/ Section: A  
PL/SQL procedure successfully completed.  
SQL> UPDATE employees  
2 SET salary=salary+1000  
3 WHERE employee_id=106;  
  
1 row updated.  
  
SQL>  
SQL> COMMIT;  
  
Commit complete.
```

Step 4: Second update

```
Name: Suresh Dahal/ Roll: 23/ Section: A  
PL/SQL procedure successfully completed.  
SQL> UPDATE employees  
2 SET salary=salary+500  
3 WHERE employee_id=106;  
  
1 row updated.  
  
SQL>  
SQL> COMMIT;  
  
Commit complete.
```

Step 5: Delete the row

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> DELETE
  2  FROM employees
  3  WHERE employee_id=106;

1 row deleted.

SQL>
SQL> COMMIT;

Commit complete.
```

Step 6: Display all versions

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> SELECT
  2      versions_starttime,
  3      versions_operation,
  4      versions_xid,
  5      salary
  6  FROM employees
  7  VERSIONS BETWEEN TIMESTAMP
  8  SYSTIMESTAMP-INTERVAL '15' MINUTE
  9  AND SYSTIMESTAMP
 10  WHERE employee_id=106;

VERSIONS_STARTTIME          V VERSIONS_XID          SALARY
-----
01-AUG-26 08.32.54 PM      D 0A000B00B7050000      6300
01-AUG-26 08.30.35 PM      U 09000B00B0020000      6300
01-AUG-26 08.25.34 PM      U 0A001500B6050000      5800
                                4800
```

(c) Analysis of Version Query Results

Step 1: Identify the transaction ID

```
VERSIONS_XID
-----
0A000B00B7050000
09000B00B0020000
0A001500B6050000
```

Step 2: Restore the deleted row

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> INSERT INTO employees
  2  SELECT *
  3  FROM employees
  4  AS OF TIMESTAMP
  5  (
  6      SYSTIMESTAMP-INTERVAL '5' MINUTE
  7  )
  8  WHERE employee_id=106;

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

Step 3: Verify restoration

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> select first_name, last_name from employees where employee_id = 106;

FIRST_NAME          LAST_NAME
-----
Valli                Pataballa
```

(d) Flashback Transaction Query Setup and Execution

Step 1: Perform a transaction

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> UPDATE employees
  2  SET salary=salary+1000
  3  WHERE employee_id=107;

1 row updated.

SQL> exec my_info;
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL>
SQL> DELETE
  2  FROM employees
  3  WHERE employee_id=206;

1 row deleted.
```

Step 2: Obtain the transaction ID before committing

```
Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

SQL> SELECT RAWTOHEX(xid) AS xid
  2  FROM v$transaction;

XID
-----
02001E009B020000
```

Step 3: Commit the transaction

```
SQL> commit;

Commit complete.
```

Step 4: Query Flashback Transaction Query

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> SELECT
  2      operation,
  3      table_name,
  4      undo_sql
  5  FROM flashback_transaction_query
  6  WHERE xid=HEXTORAW('08001B00A8020000');
```

OPERATION	TABLE_NAME	UNDO_SQL
UPDATE	EMPLOYEES	update "HR"."EMPLOYEES" set "SALARY" = '6200' where ROWID = 'AAARywAAEAAAIwPAAR';
UPDATE	EMPLOYEES	update "HR"."EMPLOYEES" set "SALARY" = '9000' where ROWID = 'AAARywAAEAAAIwPAAT';
UPDATE	EMPLOYEES	update "HR"."EMPLOYEES" set "SALARY" = '5200' where ROWID = 'AAARywAAEAAAIwPAAR';

(e) Reversing Changes with Transaction Query

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> update "HR"."EMPLOYEES" set "SALARY" = '6200' where ROWID = 'AAARywAAEAAAIwPAAR';
1 row updated.
```

```
SQL> update "HR"."EMPLOYEES" set "SALARY" = '9000' where ROWID = 'AAARywAAEAAAIwPAAT';
1 row updated.
```

Verify the changes

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> select first_name, last_name, salary from employees where employee_id in (107, 108);
```

FIRST_NAME	LAST_NAME	SALARY
Diana	Lorentz	6200
Nancy	Greenberg	12008

(f) Validation and Advanced Scenario

Step 1: Attempt a Version Query beyond available undo history

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> SELECT *
  2  FROM employees
  3  AS OF TIMESTAMP
  4  (
  5      SYSTIMESTAMP-INTERVAL '2' MONTH
  6  )
  7  WHERE employee_id=106;
FROM employees
  *
```

ERROR at line 2:
ORA-08180: no snapshot found based on specified time

Step 2: Perform a larger transaction

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> UPDATE employees
  2 SET salary=salary+500
  3 WHERE employee_id=109;
```

1 row updated.

```
SQL>
SQL> UPDATE departments
  2 SET manager_id=200
  3 WHERE department_id=20;
```

1 row updated.

Step 3: Obtain the transaction ID

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> SELECT RAWTOHEX(xid) AS xid
  2 FROM v$transaction;
```

```
XID
-----
0A001000B7050000
```

Step 4: Commit

```
SQL> commit;
```

Commit complete.

Step 5: Query transaction information

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> SELECT
  2     operation,
  3     table_name,
  4     undo_sql
  5 FROM flashback_transaction_query
  6 WHERE xid=HEXTORAW('0A001000B7050000');
```

OPERATION	TABLE_NAME	UNDO_SQL
UPDATE	DEPARTMENT	update "HR"."DEPARTMENTS" set "MANAGER_ID" = '201' where ROWID = 'AAARywAAEAAAIvtAAB';
UPDATE	EMPLOYEES	update "HR"."EMPLOYEES" set "SALARY" = '9000' where ROWID = 'AAARywAAEAAAIwPAAT';
UPDATE	EMPLOYEES	update "HR"."EMPLOYEES" set "SALARY" = '8900' where ROWID = 'AAARywAAEAAAIwPAAT';
UPDATE	EMPLOYEES	update "HR"."EMPLOYEES" set "SALARY" = '7200' where ROWID = 'AAARywAAEAAAIwPAAR';
BEGIN		

Step 6: Execute the generated UNDO_SQL statements

```
SQL> update "HR"."EMPLOYEES" set "SALARY" = '9000' where ROWID = 'AAARywAAEAAAIwPAAT';  
1 row updated.  
  
SQL> update "HR"."EMPLOYEES" set "SALARY" = '8900' where ROWID = 'AAARywAAEAAAIwPAAT';  
1 row updated.
```

Step 7: Verify

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> SELECT employee_id,  
2          salary  
3 FROM employees  
4 WHERE employee_id=109;
```

EMPLOYEE_ID	SALARY
109	8900

```
SQL>  
SQL> SELECT department_id,  
2          manager_id  
3 FROM departments  
4 WHERE department_id=20;
```

DEPARTMENT_ID	MANAGER_ID
20	200

16. AUTOMATIC WORKLOAD REPOSITORY (AWR) IN ORACLE 19C

OUTPUT

(a) Preparation and Configuration

Step 1: Connect to Oracle as SYSDBA

```
C:\Users\VICTUS>sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Sat Aug 1 21:27:37 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

Step 2: Verify AWR parameters

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT name, value
       2 FROM v$parameter
       3 WHERE name IN
       4 ('statistics_level',
       5 'control_management_pack_access');

NAME                                VALUE
-----                                -
statistics_level                    TYPICAL
control_management_pack_access      DIAGNOSTIC
+TUNING
```

Step 3: Modify parameters

```
SQL> ALTER SYSTEM SET statistics_level='TYPICAL';

System altered.

SQL>
SQL> ALTER SYSTEM SET control_management_pack_access='DIAGNOSTIC+TUNING';

System altered.
```

Step 4: Check the default snapshot interval

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT
       2 extract(day FROM snap_interval)*24*60+
       3 extract(hour FROM snap_interval)*60+
       4 extract(minute FROM snap_interval)
       5 AS snapshot_interval
       6 FROM dba_hist_wr_control;

SNAPSHOT_INTERVAL
-----
60
```

(b) Create Baselines with Snapshots

Step 1: Create the first AWR snapshot

```
Name: Suresh Dahal/Roll: 23/Section: A
PL/SQL procedure successfully completed.
SQL> EXEC DBMS_WORKLOAD_REPOSITORY.CREATE_SNAPSHOT;
PL/SQL procedure successfully completed.
```

Step 2: Connect as HR

```
SQL> connect hr/thaxaena@orclpdb;
Connected.
```

Step 3: Generate workload

```
Name: Suresh Dahal/ Roll: 23/ Section: A
PL/SQL procedure successfully completed.
SQL> SELECT e.employee_id,
2         e.first_name,
3         d.department_name
4 FROM employees e
5 JOIN departments d
6 ON e.department_id=d.department_id;

EMPLOYEE_ID FIRST_NAME          DEPARTMENT_NAME
-----
200 Jennifer          Administration
201 Michael            Marketing
202 Pat                Marketing
114 Den                Purchasing
119 Karen              Purchasing
115 Alexander           Purchasing
116 Shelli              Purchasing
117 Sigal                Purchasing
118 Guy                 Purchasing
```

```
Name: Suresh Dahal/ Roll: 23/ Section: A
PL/SQL procedure successfully completed.
SQL> SELECT d.department_name,
2         COUNT(*)
3 FROM employees e
4 JOIN departments d
5 ON e.department_id=d.department_id
6 GROUP BY d.department_name;

DEPARTMENT_NAME          COUNT(*)
-----
Sales                     34
Marketing                  2
Administration             1
Purchasing                 6
Shipping                   45
IT                          4
Executive                  3
Finance                    6
Public Relations           1
Human Resources             1
Accounting                  1

11 rows selected.
```

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> SELECT employee_id,  
2         first_name,  
3         salary  
4 FROM employees  
5 WHERE salary >  
6 (  
7 SELECT AVG(salary)  
8 FROM employees  
9 );
```

EMPLOYEE_ID	FIRST_NAME	SALARY
201	Michael	13000
203	Susan	6500
204	Hermann	10000
205	Shelley	12008
100	Steven	24000
101	Neena	17000
102	Lex	17000
103	Alexander	8000

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> SELECT department_id,  
2         AVG(salary)  
3 FROM employees  
4 GROUP BY department_id;
```

DEPARTMENT_ID	AVG(SALARY)
50	3475.55556
40	6500
110	12008
70	10000
90	19333.3333
30	4150
10	4400
	7000
20	9500
60	6225
100	8584.66667

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> UPDATE employees  
2 SET salary = salary + 100  
3 WHERE employee_id = 101;
```

1 row updated.

SQL>

SQL> COMMIT;

Commit complete.

Restore the original salary.

Name: Suresh Dahal/ Roll: 23/ Section: A

PL/SQL procedure successfully completed.

```
SQL> UPDATE employees  
2 SET salary = salary - 100  
3 WHERE employee_id = 101;
```

1 row updated.

Step 4: Create the second snapshot

```
SQL> connect / as sysdba;
Connected.
SQL> EXEC DBMS_WORKLOAD_REPOSITORY.CREATE_SNAPSHOT;

PL/SQL procedure successfully completed.
```

Step 5: View available snapshots

Display all snapshots.

```
Name: Suresh Dahal/Roll: 23/Section: A

PL/SQL procedure successfully completed.

SQL> SELECT snap_id,
2         begin_interval_time,
3         end_interval_time
4 FROM dba_hist_snapshot
5 ORDER BY snap_id;
```

SNAP_ID	BEGIN_INTERVAL_TIME	END_INTERVAL_TIME
375	24-JUL-26 06.46.04.429 AM	24-JUL-26 07.45.14.847 AM
376	24-JUL-26 07.45.14.847 AM	24-JUL-26 03.46.39.359 PM
377	24-JUL-26 03.46.39.359 PM	24-JUL-26 04.45.47.765 PM
378	24-JUL-26 04.45.47.765 PM	24-JUL-26 05.45.56.127 PM
379	24-JUL-26 05.45.56.127 PM	24-JUL-26 06.53.29.798 PM
380	24-JUL-26 06.53.29.798 PM	24-JUL-26 07.45.38.257 PM
381	24-JUL-26 07.45.38.257 PM	24-JUL-26 08.45.47.982 PM
382	24-JUL-26 08.45.47.982 PM	24-JUL-26 09.45.48.338 PM
383	24-JUL-26 09.45.48.338 PM	24-JUL-26 10.45.58.644 PM
384	24-JUL-26 10.45.58.644 PM	25-JUL-26 08.32.28.076 AM
385	25-JUL-26 08.32.28.076 AM	25-JUL-26 09.45.42.514 AM
386	25-JUL-26 09.45.42.514 AM	25-JUL-26 10.45.12.481 AM
387	25-JUL-26 10.45.12.481 AM	25-JUL-26 11.45.21.614 AM
388	25-JUL-26 11.45.21.614 AM	25-JUL-26 12.45.21.001 PM
389	25-JUL-26 12.45.21.001 PM	25-JUL-26 01.45.29.287 PM
390	25-JUL-26 01.45.29.287 PM	25-JUL-26 02.45.07.016 PM
477	01-AUG-26 04.45.38.636 PM	01-AUG-26 05.45.47.058 PM
478	01-AUG-26 05.45.47.058 PM	01-AUG-26 06.45.38.331 PM
479	01-AUG-26 06.45.38.331 PM	01-AUG-26 07.45.06.083 PM
480	01-AUG-26 07.45.06.083 PM	01-AUG-26 08.45.15.471 PM
481	01-AUG-26 08.45.15.471 PM	01-AUG-26 09.31.15.785 PM
482	01-AUG-26 09.31.15.785 PM	01-AUG-26 09.38.07.050 PM

(c) Generate AWR Report

Step 1: Run the AWR report script

```
SQL> @?/rdbms/admin/awrrpt.sql

Specify the Report Type
*****
AWR reports can be generated in the following formats. Please enter the
name of the format at the prompt. Default value is 'html'.

'html'          HTML format (default)
'text'          Text format
'active-html'   Includes Performance Hub active report
```

Step 2: Select report type

```
SQL> @?/rdbms/admin/awrrpt.sql

Specify the Report Type
~~~~~
AWR reports can be generated in the following formats. Please enter the
name of the format at the prompt. Default value is 'html'.

'html'          HTML format (default)
'text'          Text format
'active-html'   Includes Performance Hub active report

Enter value for report_type: html
old  1: select 'Type Specified: ',lower(nvl('&&report_type','html')) report_type from dual
new  1: select 'Type Specified: ',lower(nvl('html','html')) report_type from dual

Type Specified:  html
```

Step 3: Specify the number of days

```
Enter value for num_days: 1

Listing the last day's Completed Snapshots
Instance      DB Name      Snap Id      Snap Started      Snap Level
-----
orcl          ORCL          467 01 Aug 2026 00:45 1
              468 01 Aug 2026 07:07 1
              469 01 Aug 2026 07:45 1
              470 01 Aug 2026 09:52 1
              471 01 Aug 2026 10:45 1
              472 01 Aug 2026 12:36 1
              473 01 Aug 2026 13:45 1
              474 01 Aug 2026 14:45 1
              475 01 Aug 2026 15:45 1
              476 01 Aug 2026 16:45 1
              477 01 Aug 2026 17:45 1
              478 01 Aug 2026 18:45 1
              479 01 Aug 2026 19:45 1
              480 01 Aug 2026 20:45 1
              481 01 Aug 2026 21:31 1
              482 01 Aug 2026 21:38 1
```

Step 4: Enter Begin Snapshot ID

```
Specify the Begin and End Snapshot Ids
~~~~~
Enter value for begin_snap: 481
Begin Snapshot Id specified: 481
```

Step 5: Enter End Snapshot ID

```
Enter value for end_snap: 482
End Snapshot Id specified: 482
```

Step 6: Specify the report name

```
Specify the Report Name
~~~~~
The default report file name is awrrpt_1_481_482.html. To use this name,
press <return> to continue, otherwise enter an alternative.

Enter value for report_name: awr_report.html

Using the report name awr_report.html
```

Step 7: Open the generated report

WORKLOAD REPOSITORY report for

DB Name	DB Id	Unique Name	Role	Edition	Release	RAC	CDB
ORCL	1764205357	orcl	PRIMARY	EE	19.0.0.0.0	NO	YES

Instance	Inst Num	Startup Time
orcl	1	01-Aug-26 12:25

Container DB Id	Container Name	Open Time
1764205357	CDB\$ROOT	01-Aug-26 12:25
1793743613	PDB\$SEED	01-Aug-26 12:25
3460620442	ORCLPDB	01-Aug-26 12:53

Host Name	Platform	CPUs	Cores	Sockets	Memory (GB)
ACONITIN	Microsoft Windows x86 64-bit	12	6	1	15.23

Snap Id	Snap Time	Sessions	Cursors/Session	Pluggable Databases Open
Begin Snap: 481	01-Aug-26 21:31:15	61	1.3	3
End Snap: 482	01-Aug-26 21:38:07	61	1.3	3
Elapsed:	6.85 (mins)			
DB Time:	0.08 (mins)			

(d) Analysis of AWR Report Sections

1. Instance Efficiency Percentages

Instance Efficiency Percentages (Target 100%)

Buffer Nowait %:	100.00	Redo NoWait %:	100.00
Buffer Hit %:	99.75	In-memory Sort %:	100.00
Library Hit %:	88.47	Soft Parse %:	72.96
Execute to Parse %:	61.37	Latch Hit %:	99.87
Parse CPU to Parse Elapsed %:	78.13	% Non-Parse CPU:	96.14
Flash Cache Hit %:	0.00		

2. Top Timed Foreground Events

Top 10 Foreground Events by Total Wait Time

Event	Waits	Total Wait Time (sec)	Avg Wait	% DB time	Wait Class
Failed Logon Delay	3	3	1014.45ms	66.6	Other
DB CPU		1.4		30.5	
Disk file operations I/O	33	.1	2.23ms	1.6	User I/O
control file sequential read	244	0	119.35us	.6	System I/O
db file sequential read	78	0	287.86us	.5	User I/O
log file sync	5	0	944.00us	.1	Commit
PGA memory operation	240	0	7.87us	.0	Other
PX Deq: Join ACK	2	0	681.50us	.0	Other
PX Deq: Slave Session Stats	4	0	183.00us	.0	Other
direct path write	10	0	50.10us	.0	User I/O

3. SQL Statistics

SQL ordered by Elapsed Time

- Resources reported for PL/SQL code includes the resources used by all SQL statements called by the code.
- % Total DB Time is the Elapsed Time of the SQL statement divided into the Total Database Time multiplied by 100
- %Total - Elapsed Time as a percentage of Total DB time
- %CPU - CPU Time as a percentage of Elapsed Time
- %IO - User I/O Time as a percentage of Elapsed Time
- Captured SQL account for 49.2% of Total DB Time (s): 5
- Captured PL/SQL account for 50.3% of Total DB Time (s): 5

Elapsed Time (s)	Executions	Elapsed Time per Exec (s)	%Total	%CPU	%IO	SQL Id	SQL Module	PDB Name	SQL Text
1.25	1	1.25	27.28	91.40	5.60	491wcfyf6wc1	sqlplus.exe	CDB\$ROOT	BEGIN DBMS_WORKLOAD_REPOSITORY...
0.73	1	0.73	16.02	94.91	0.84	fhf8upax5cxsz		CDB\$ROOT	BEGIN sys.dbms_auto_report_int...
0.70	1	0.70	15.22	93.34	0.88	0w26sk6t6gq98		CDB\$ROOT	SELECT XMLTYPE(DBMS_REPORT.GET...
0.63	1	0.63	13.73	93.15	0.98	9yv5dwv8k0awg		CDB\$ROOT	WITH MONITOR_DATA AS (SELECT I...
0.29	1	0.29	6.26	100.04	0.59	6ajkhukk78nsr		CDB\$ROOT	begin prvt_hdm.auto_execute(:...
0.13	3	0.04	2.85	85.43	1.20	dvu40a9avazf8		CDB\$ROOT	select xmlagg(xmlelement("ope...
0.12	3	0.04	2.53	87.15	2.33	3rdvj2msrkpn0		CDB\$ROOT	select T.* from table(dbms_xpl...
0.06	1	0.06	1.40	18.23	67.34	dyycftbp20dc7		CDB\$ROOT	insert into wrh\$_tablespace_st...
0.05	644	0.00	1.07	85.30	31.09	f3ww8rgva3hrs		CDB\$ROOT	update /* KSM:FLUSH COL */ sy...
0.05	1	0.05	1.00	93.20	4.75	fg91fmh6ux23c		CDB\$ROOT	insert into wrh\$_sga_target_ad...

4. Performance Analysis and Optimization

Instance Efficiency Percentages (Target 100%)

Buffer Nowait %:	100.00	Redo NoWait %:	100.00
Buffer Hit %:	99.75	In-memory Sort %:	100.00
Library Hit %:	88.47	Soft Parse %:	72.96
Execute to Parse %:	61.37	Latch Hit %:	99.87
Parse CPU to Parse Elapsed %:	78.13	% Non-Parse CPU:	96.14
Flash Cache Hit %:	0.00		